Project 14: Ravenhill Coffee Shop Management System

About the Client

Ravenhill Coffee is a specialty coffee roaster and retailer, dedicated to sourcing, roasting and sharing exceptional coffee in a sustainable, respectful and responsible way. We want to establish our first coffee shop in Melbourne CBD starting January 2023 and we need our own POS system to manage our shop.

Ravenhill Coffee is a specialty coffee roaster and retailer, born and based in Melbourne. We are dedicated to sourcing, roasting and making beautiful coffee, and we're committed to doing so in a responsible, sustainable and respectful way.

Ravenhill was founded in 2009 by Fleur Studd and Jason Scheltus. It's hard to fathom now, but at that time, finding fresh, in-season, traceable, high quality coffee in Melbourne (or Australia) was practically impossible. We wanted to help address that and, in doing so, ignite positive change in the industry by redefining what coffee was and could be, and to build a market for, and appreciation of, high quality specialty coffee.

We built our first shop and roastery in the wonderful, bustling Prahran Market. We fell in love with this location for many reasons, the most important being that it allows us to engage with a community of shoppers who are seeking out quality produce, and who care about seasonality and provenance.

We realised early on that we wanted to deliver Melbourne's best coffee experience and to do this, we needed to put all our energy into sourcing, roasting and sharing the very best coffee we could find. Our goal was to make these coffees accessible and exciting, easy to understand and appreciate, and simple to brew and enjoy.

Project Brief & Business Problem Specifications:

This project should provide an automated platform for coffee shop, so that Ravenhill Coffee could handle their daily sales transactions in a more efficient and productive manner. The application should provide the user with the capability to save, manage, and retrieve the sales transactions associated with the specified business. The management of the Ravenhill Coffee Shop is the only ones who have access to this POS for their shop. It is necessary for the

management to log in to the system using their current valid system credentials in order for them to be granted access to the features and functions that are available. The Administrator, the Staff, and the Cashier are the three distinct sorts of user roles that are able to log in and use this system. The Admin user should have the ability to access and administer all of the features and operations of the system, including the management of users and system information. This capability should only available to the Admin user. The Cashier users should only access the sales transactions that they themselves have generated, whereas the Staff users should have restricted capabilities. Additionally, the project should provide a daily sales report as well as a printable receipt.

The user of the system must provide his or her user credential in order to access and use the features and functionalities of this straightforward project because it will possess a secure login feature. This means that the user can only access and use the features and functionalities of this straightforward project. In addition to storing the sales transactions, the system should also have the capability of storing a list of the many types of water jars and containers, as well as their prices.

System Modules

1. User Management Module

- Customer registration and login
- Staff and admin account management
- Password reset and recovery
- Profile editing (name, email, address)
- Role-based access control

2. Product Catalogue Module

- Add, edit, remove coffee items and other menu products
- Upload product images and descriptions
- Organise items into categories (e.g., Hot Drinks, Cold Drinks, Pastries)
- Set prices and availability
- Highlight featured items

3. Online Ordering Module

- Browse menu items with images and details
- Customise orders (size, milk type, extras)
- Add items to shopping cart
- Review and confirm orders
- Save favourite orders for reordering

4. Shopping Cart and Checkout Module

- Add/remove items from cart
- Show order summary and total price
- Apply promotional codes or discounts
- Select pickup or delivery options
- Secure payment processing

5. Payment Management Module

- Support multiple payment methods (card, wallet, cash on delivery)
- Process payments securely
- Generate digital receipts
- Handle refunds or adjustments
- Store payment history

6. Order Management Module

- Store and track orders with status (Pending, In Progress, Ready, Completed)
- Assign orders to staff/baristas
- Update order status in real-time
- Notify customers about status changes
- View order history

7 . Inventory Management Module

- Track stock levels for ingredients and packaging
- Record stock in/out transactions
- Generate low-stock alerts
- Manage supplier details
- View inventory usage reports

9. Loyalty and Rewards Module

- Track loyalty points for customers
- Define rewards or discount tiers
- Apply discounts based on loyalty status
- View loyalty activity history
- Notify customers of earned rewards

10. Promotions and Discounts Module

- Create and manage promo codes
- Schedule time-limited discounts
- Apply automatic discounts to eligible orders
- Track promotion usage and effectiveness

11. Notifications and Communication Module

- Send order confirmations via email/SMS
- Notify customers about order status updates
- Send marketing emails or offers
- Allow internal messages between staff roles

12. Reviews and Feedback Module

- Collect customer reviews on products or orders
- Display average ratings on products

- Enable text feedback
- Moderate and manage reviews

13. Reporting and Analytics Module

- Daily/weekly/monthly sales summaries
- Best-selling products analysis
- Order volume trends
- Inventory usage and wastage reports
- Customer activity and loyalty insights

14. Admin Dashboard Module

- Manage user accounts and roles
- Oversee products and pricing
- View and manage orders
- Access sales and inventory reports
- Configure system settings

15. Content Management Module

- Manage static site pages (About, Contact, FAQs)
- Update store information (hours, location, delivery areas)
- Upload promotional banners or images
- Edit SEO settings

User Modules (User Frontend):

Developers need to research and discuss with the client to finalise the modules and requirements.

Input Data and Validation Requirements

- Avoiding errors in data
- Controlling amount of input
- Integration of all the modules/forms in the system.
- Preparation of the test cases.
- Preparation of the possible test data with all the validation checks.
- Black-box/White-box testing.
- Recording of all the reproduced errors.
- Modifications should be done for the errors found during testing.
- Prepare the test result scripts after rectification of the errors
- Functionality of the entire modules/forms.
- Validations for user input.
- Checking of the Coding standards to be maintained during coding.
- Testing the modules with all the possible test data.
- Testing of the functionality involving all type of calculations.
- Commenting standard in the source files.
- Easy & fast retrieval of information.
- Well-designed reports.

- Decrease the load of the person involve in existing manual system.
- Access of any information individually.
- Work become very speedy.
- Easy to update information

UI Design

User Interface Design is concerned with the dialogue between a user and the computer. It is concerned with everything from starting the system or logging into the system to the eventually presentation of desired inputs and outputs. The overall flow of screens and messages is called a dialogue.

UI Design Requirements

1. The system user should always be aware of what to do next.
2. The screen should be formatted so that various types of information, instructions and messages always appear in the same general display area.
3. Message, instructions or information should be displayed long enough to allow the system user to read them.
4. Use display attributes sparingly.
5. Default values for fields and answers to be entered by the user should be specified.
6. A user should not be allowed to proceed without correcting an error.
7. The system user should never get an operating system message or fatal error.

The aim of proposed system is to develop a system of improved facilities.

The proposed system can overcome all the limitations of the existing system.

The system provides proper security and reduces the manual work.

- Security of data.
- Ensure data accuracy's
- Proper control of the higher officials.
- Minimize manual data entry.
- Minimum time needed for the various processing.
- Greater efficiency.
- Better service.
- User friendliness and interactive.
- Minimum time required.

Functional Requirements

FR1. Allow customer registration with email verification.
FR2. Provide secure login and logout functionality.
FR3. Enable password reset and account recovery.
FR4. Let users edit personal profile information.
FR5. Support role-based access for customers, staff, and administrators.
FR6. Add new coffee products with name, description, price, and category.
FR7. Upload and manage product images.
FR8. Edit or remove existing menu items.
FR9. Categorise products into groups like Hot Drinks, Cold Drinks, and Pastries.
FR10. Set product availability status.
FR11. Display menu items with images, prices, and descriptions.
FR12. Allow customers to customise orders (size, milk type, extras).
FR13. Enable adding items to a shopping cart.
FR14. Support reviewing and confirming orders before checkout.
FR15. Provide ability to save favourite orders for quick reordering.
FR16. Enable adding and removing items from the shopping cart.
FR17. Display real-time order total in the cart.
FR18. Apply discount codes or promotions during checkout.
FR19. Allow customers to choose pickup or delivery options.
FR20. Process secure payments through supported methods.
FR21. Record orders with status tracking (Pending, In Progress, Ready, Completed).
FR22. Let staff view and manage active orders.
FR23. Update order status in real-time.

FR24. Notify customers about order status changes.
FR25. Allow customers to view their order history.
FR26. Manage delivery addresses for customers.
FR27. Assign delivery orders to drivers.
FR28. Track and update delivery status.
FR29. Notify customers when orders are out for delivery.
FR30. Confirm delivery completion with timestamp.
FR31. Track stock levels for ingredients and packaging materials.
FR32. Record inventory in and out transactions.
FR33. Generate low-stock alerts for managers.
FR34. Manage supplier details for ordering supplies.
FR35. Produce inventory usage and wastage reports.
FR36. Track loyalty points for customers.
FR37. Define and manage reward tiers or discount levels.
FR38. Automatically apply loyalty discounts during checkout.
FR39. Allow customers to view their loyalty history.
FR40. Notify customers about earned rewards or promotions.
FR41. Create, edit, and remove promotional codes.
FR42. Define discount rules such as percentage or fixed amount.
FR43. Schedule time-limited discounts or offers.
FR44. Apply promotions automatically to eligible orders.
FR45. Track and report promotion usage.
FR46. Send email or SMS order confirmations to customers.
FR47. Notify customers about changes in order status.
FR48. Send marketing emails for promotions and updates.
FR49. Support internal messaging between staff roles.
FR50. Allow customers to set notification preferences.
FR51. Enable customers to submit reviews for products.
FR52. Store reviews and ratings in the database.
FR53. Display average product ratings on product pages.
FR54. Allow written text reviews with moderation capability.
FR55. Let admins approve, edit, or delete inappropriate reviews.
FR56. Generate sales reports by day, week, or month.
FR57. Summarise total revenue and payment methods used.
FR58. Report on best-selling products.
FR59. Track order volume trends over time.
FR60. Produce inventory usage and wastage reports.
FR61. Provide admin dashboard with user, order, and sales statistics.
FR62. Allow management of user accounts and roles.
FR63. Enable adding, editing, and removing products.
FR64. Review and manage all orders and payments.
FR65. Access reports and export data to CSV or PDF formats.
FR66. Manage static site content such as About Us, Contact, and FAQs pages.
FR67. Update store details like opening hours, address, and delivery areas.

FR68. Upload promotional banners or images.
FR69. Edit SEO metadata for website optimisation.
FR70. Provide easy content updates via admin panel.
FR71. Support responsive design for desktops, tablets, and mobile devices.
FR72. Provide consistent navigation and UI across devices.
FR73. Ensure role-based access controls work on mobile interfaces.
FR74. Log and track mobile app usage activity.

Non-functional Requirements

Apart from other NFRs following NFRs must be implemented in the system:

Less clicks and keyboard interactions
Consistent look between website/app and the style in store
Account/phone number verification

Hardware Requirement: Should be recommended by the developers.

Software Requirement: Should be recommended by the developers.